CSC 4260/5260 Team Project Spring 2026

DESCRIPTION

The team projects for this class are to be done in **groups of 4-5 members.**

The idea is to experience the complete data science pipeline (data gathering, cleaning, modeling, interpretation, communication).

The following is a list of potential projects your team can choose from. Your team will specify your preferences either from among these projects, or you can propose a different theme. Teams will be assigned to projects after reviewing each team's submitted prioritized list.

PROJECT IDEAS/THEMES

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1. Education / Learning Sciences

- **Possible Question**: What factors best predict whether a student will persist to the next semester or graduate on time? What features of online course activity (clickstream, submission times, discussion forum use) predict student performance or retention? Can we detect "at-risk" students earlier using patterns in gradebook and LMS data? How does course scheduling (time of day, sequence of prerequisites) impact student outcomes?
- **Possible Data Sources**: Institutional data (GPA, courses taken, financial aid, advising visits), student engagement (clubs, LMS activity).
- **Methods**: Classification (logistic regression, random forest), feature importance analysis, survival analysis.
- ***Note**: I have a graduate student currently working on a similar problem who could be a resource.*

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2. Campus Operations / Student Life

- **Possible Questions**: Can we predict traffic and food demand in dining halls to reduce waste and improve service? What factors drive student use of university shuttle/bus systems, and how could routing be optimized? How do student event attendance patterns relate to retention and engagement?
- **Possible Data Sources**: Dining swipe card data, menus, time/date, weather, major events (sports, concerts), bus/shuttle schedules and routes.
- **Methods**: Time series forecasting (ARIMA, Prophet), regression, anomaly detection.

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3. Health Sciences / Nursing / Public Health

- **Possible Questions**: How do self-reported health and lifestyle factors (sleep, exercise, diet, stress) affect academic performance? Can patterns in student health center visits predict peak demand periods for certain services? Do local environmental factors (e.g., air quality, pollen counts, weather) affect reported student health issues?

- **Possible Data Sources**: Anonymous student survey, GPA/course data, fitness center usage logs, health center data, weather data.
- **Methods**: Regression modeling, clustering (to identify "lifestyle profiles"), correlation analysis.

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4. Sustainability / Environmental Science

- **Possible Questions**: What are the biggest drivers of campus energy use and waste production, and how can they be reduced? Can satellite or drone imagery be used to track and predict changes in campus green spaces? What are the largest contributors to the university's carbon footprint, and how can data-driven strategies reduce them?
- **Possible Data Sources**: IoT sensor data (building energy meters, recycling/waste bins), class schedules, weather data, drone data.
- **Methods**: Time series analysis, regression, geospatial mapping, clustering.

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5. Athletic Performance and Injury Prediction

- **Possible Questions**: Can we predict injury risk or performance outcomes based on training and biometric data? How do training schedules, sleep patterns, and travel affect athletic performance? What role does game-day weather play in performance outcomes for outdoor sports?
- **Possible Data Sources**: Team practice schedules, wearable fitness trackers, game-day stats, weather conditions.
- **Methods**: Predictive modeling (classification/regression), survival analysis, feature selection.

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6. Course Design and Student Outcomes

- **Possible Questions**: How do factors like course timing, sequence, and modality (online/in-person) impact student performance?
- **Possible Data Sources**: Registrar's office data (course schedules, grades, modality), demographics.
- **Methods**: Regression modeling, causal inference (propensity score matching), visualization.

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7. Arts / Humanities / Social Sciences

- **Possible Questions**: What linguistic patterns distinguish student writing in STEM vs. humanities courses? Can sentiment analysis of campus newspaper articles or student surveys detect changes in campus climate? How do patterns of social media engagement differ across different student clubs and organizations?
- **Possible Data Sources**: Anonymized student essays, lab reports, discussion posts, newspapers, social media.
- **Methods**: NLP (topic modeling, sentiment analysis, word embeddings), classification.

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8. Engineering / Computer Science

- **Possible Questions**: Can IoT sensor data be used to optimize energy usage? How well do different machine learning algorithms perform in predicting outcomes from engineering experiment data?

Can GitHub commit history and project communication logs predict the success of student team projects?

- **Possible Data Sources**: IoT data from smart labs or campus buildings (e.g., AIEB), CS/engineering lab data, git hub logs.
- **Methods**: Geospatial analysis, clustering (stop demand), optimization models.

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9. Business / Economics / Finance

- **Possible Questions**: Can we predict credit risk or loan repayment likelihood using anonymized student financial aid data? What are the most influential factors in determining student startup success in university incubators? How do local economic indicators affect enrollment trends?
- **Possible Data Sources**: Student financial data, Biz Foundry, Cookeville Chamber of Commerce.
- **Methods**: Time series forecasting (ARIMA, Prophet), regression.

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10. Cross-Disciplinary Research Productivity

- **Possible Question**: What factors are most predictive of student/faculty research productivity (presentations, publications, funding)?
- **Possible Data Sources**: Institutional research office (grant data, faculty-student research projects), publication databases, student demographics.
- **Methods**: Regression modeling, feature importance, clustering (to find research "profiles").

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11. Campus Scientific Projects

The following are projects on campus that faculty ***might*** be willing to work with you on collecting the data, asking relevant questions, and guiding/mentoring you on your project:

- **Topic***: Precision Agriculture for Rural Empowerment (Dr. Abdul Momim)*
  Potential Subject*: Utilize remote sensing data for crop health and growth monitoring and historical agricultural data for trend analysis.*

- **Topic***: Environmental Justice Patterns in Rural Tennessee (Dr. Peter Li)*
  Potential Subject*: Identify patterns in spatial data.*

- **Topic***: Development of flood forecasting models for rural watersheds (Dr. Alfred Kalyanapu)*
  Potential Subject*: Develop ML and AI-based flood forecasting models for rural watersheds.*

- **Topic***: Quantifying waterfowl population numbers using aerial UAV imagery (Dr. Bradley Cohen)*
  Potential Subject*: Use machine learning algorithms to recognize, identify to species, identify sex, and quantify the number of each subset from video and imagery captured with overhead aerial UAV.*

- **Topic***: Metabolic and Cardiovascular Health Community Risk (Dr. Melissa Geist)*
  Potential Subject*: Metabolic and cardiovascular health community risk assessment and targeted intervention with a focus on the rural and Hispanic populations in Central Appalachia.*

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12. Other

If you have an idea that does not fit into any of the themes above, you must submit the following information:

- **Theme:** What is the overall theme of the project?
- **Possible Questions**: What are some possible questions?
- **Possible Data Sources:** What data sources are you targeting?
- **Methods**: What machine learning methods/models do you believe will be needed?

DELIVERABLES

1. Project Reports (see Template on next page)
2. Link to Project Notebook/Code (GitHub), professionally documented, including
    a. Comments to better understand the code
    b. Relevant instructions to replicate results; and
    c. Who developed what?

PROJECT REPORT TEMPLATE

Your project report is the formal description of your project. Your report will be graded on: Quality, Completeness, Creativity, and Grammar.

The report should use the following template (**<u>replacing</u> the text in [*brackets*] with your text – don't include the brackets and don't use italics for your font!**)

[NAME OF PROJECT]

[TEAM MEMBER NAMES]

1. Problem Statement

[Give a clear and complete statement of the problem or question(s) you are trying to answer. (Do NOT describe data, methods or tools yet – see below.) Where does the data come from (i.e., the source of the data? What are its characteristics? Etc.) Include informal success measures (e.g., accuracy, area under the curve (AOC, RUC), precision/recall, etc.) that you plan to use.

2. Background and Related Work (<u>for CSC 5260 students only – can be in a separate document</u>)
[Include background material as appropriate: who cares about this problem, what impact it has, what implications better solutions might have. Include a scientifically professional summary of related work and the pros/cons of their proposed approaches/solutions to this type of problem.]

3. Data and Exploratory Analysis

[Describe the data set you will be using. Discuss anything you had to do to clean the data and why. Describe what tools and code you used to extract, clean, and generate the data for your experiments. Some potential questions of the data might be: any anomalies or outliers? Did you need to impute any of the data in order to get it to work for any proposed algorithms?]

4. Methods and Tools

[Describe the methods you explored (usually algorithms, or data cleaning or data wrangling approaches). Justify your methods in terms of the problem statement. What are you considering? Be sure to include every method you are trying, even if they eventually don't work. And, when describing methods that didn't work, make clear how they failed and any evaluation metrics you used to decide so. Describe the tools (e.g., libraries, packages, etc.) that you used and the reasons for their choice. Justify them in terms of the problem itself and the methods you want to use (i.e., what you indicated above). Tools will probably include machine learning, and any that were used to do data wrangling and visualization. Please discuss all of them. How did you employ them? What features worked well and what didn't? What could be improved? Describe any tools that you tried and ended up not using. What was the problem?]

5. Results

[Give a detailed summary of the results of your work. Here is where you specify the exact performance measures you used. Usually, there will be some kind of accuracy or quality measure. There may also be a performance (runtime or throughput) measure. Please use visualizations whenever possible. Include links to interactive visualizations if you built them. You should attempt to evaluate a primary model and in addition, a "baseline" model. The baseline is typically the simplest model that's applicable to that data problem, e.g., Naive Bayes for classification, or K-means on raw feature data for clustering. If there isn't a plausible automatic baseline model, you can, for example, compare with human performance by having someone hand-solve your problem on a small subset of data. You won't expect to achieve this level of performance, but it establishes a scale by which to measure your project's performance. Compare the performance of your baseline model and primary model and explain the differences.]

6. Conclusions and Future Work

[In this section, give a high-level summary of your results. If the reader only reads one section of the report, this one should be it, and it should be self-contained. You can refer back to the "Results" section for elaborations. This section should be less than a page. In particular, emphasize any results that were surprising. Finish with some concluding thoughts, including where you would go next (if you had the time)]

7. Appendix

[Include the link to your GitHub repository (that can be accessed) containing your notebook/code and link to the data. Code should be well documented (such that others could replicate your work) and professional in presentation. Also include the data, or a link to the data.]